

WORKED EXAMPLE

Newton uses slope and curvature together

Start with one parameter, where the Hessian is just a second derivative.

$$E(w) = 2(w - 1)^2, \quad g = 4(w - 1), \quad H = 4$$

$$w_0 = 3 : \quad g_0 = 8, \quad w_{new} = 3 - \frac{8}{4} = 1$$

$$E(3) = 8, \quad E(1) = 0$$

The full Newton correction divides slope by curvature. In this exact positive-curvature quadratic, it lands at the minimum in one step.

$$\text{GD at } \eta_G = 0.1 : \quad w_{new} = 3 - 0.1(8) = 2.2$$

Gradient descent uses a chosen rate times the slope. Newton adjusts the correction using curvature.

WORKED EXAMPLE

Map the lecture formula to a linear system

Evaluate the gradient and Hessian at the same original parameter vector.

$$w_{new} = w - \eta_N H^{-1} g$$

Lecture4 PDF page50 includes a rate eta. Here eta_N=1 means a full Newton step. We use g as a column gradient; tuples below list its components.

$$Hs = g, \quad w_{new} = w - s \quad (\eta_N = 1)$$

The correction s equals H-inverse times g. The actual displacement is -s. Some notes instead name the signed displacement Delta and solve H Delta = -g.

$$H\Delta = -g, \quad \Delta = -s, \quad w_{new} = w + \Delta$$

Both conventions give the same update. Solving the system avoids explicitly forming the inverse.

WORKED EXAMPLE

Compute the mixed-quadratic ingredients

Reuse the Lesson21 function and the same start for both methods.

$$E(x, y) = 1.5x^2 + xy + 1.5y^2, \quad w_0 = (2, 0)$$

$$g = (3x + y, x + 3y), \quad g_0 = (6, 2)$$

Hessian row	column x	column y
differentiate 3x+y	3	1
differentiate x+3y	1	3

$$E(2, 0) = 1.5(2)^2 + 2(0) + 1.5(0)^2 = 6$$

Thus H has rows (3,1) and (1,3). Its positive eigenvalues 2 and 4 were derived in Lesson21.

WORKED EXAMPLE

Solve for the Newton correction by elimination

The matrix equation $Hs=g$ becomes two ordinary equations.

$$3s_x + s_y = 6, \quad s_x + 3s_y = 2$$

$$s_y = 6 - 3s_x$$

$$s_x + 3(6 - 3s_x) = 2 \Rightarrow -8s_x = -16 \Rightarrow s_x = 2$$

$$s_y = 6 - 3(2) = 0$$

$$w_{new} = (2, 0) - (2, 0) = (0, 0), \quad E_{new} = 0$$

Check the solution: $3(2)+0=6$ and $2+3(0)=2$. The correction is $(2,0)$, while the displacement is $(-2,0)$.

WORKED EXAMPLE

Optional check: the two-by-two inverse

The linear solve is sufficient; this shows why the inverse formula agrees.

For rows (a,b) and (c,d), the inverse has rows (d,-b) and (-c,a), all divided by ad-bc. This requires a nonzero determinant.

$$\det H = 3(3) - 1(1) = 8$$

Therefore H inverse has rows (3,-1)/8 and (-1,3)/8.

$$s_x = \frac{3(6) - 1(2)}{8} = \frac{16}{8} = 2$$

$$s_y = \frac{-1(6) + 3(2)}{8} = \frac{0}{8} = 0$$

These are exactly the correction components found by elimination. Do not invert each matrix entry separately.

WORKED EXAMPLE

Compare gradient descent from the same start

Use $\eta_G=0.1$; this is an alternative update, not an extra Newton step.

$$w_{GD} = (2, 0) - 0.1(6, 2) = (1.4, -0.2)$$

$$\begin{aligned} E(1.4, -0.2) &= 1.5(1.4)^2 + (1.4)(-0.2) + 1.5(-0.2)^2 \\ &= 2.94 - 0.28 + 0.06 = 2.72 \end{aligned}$$

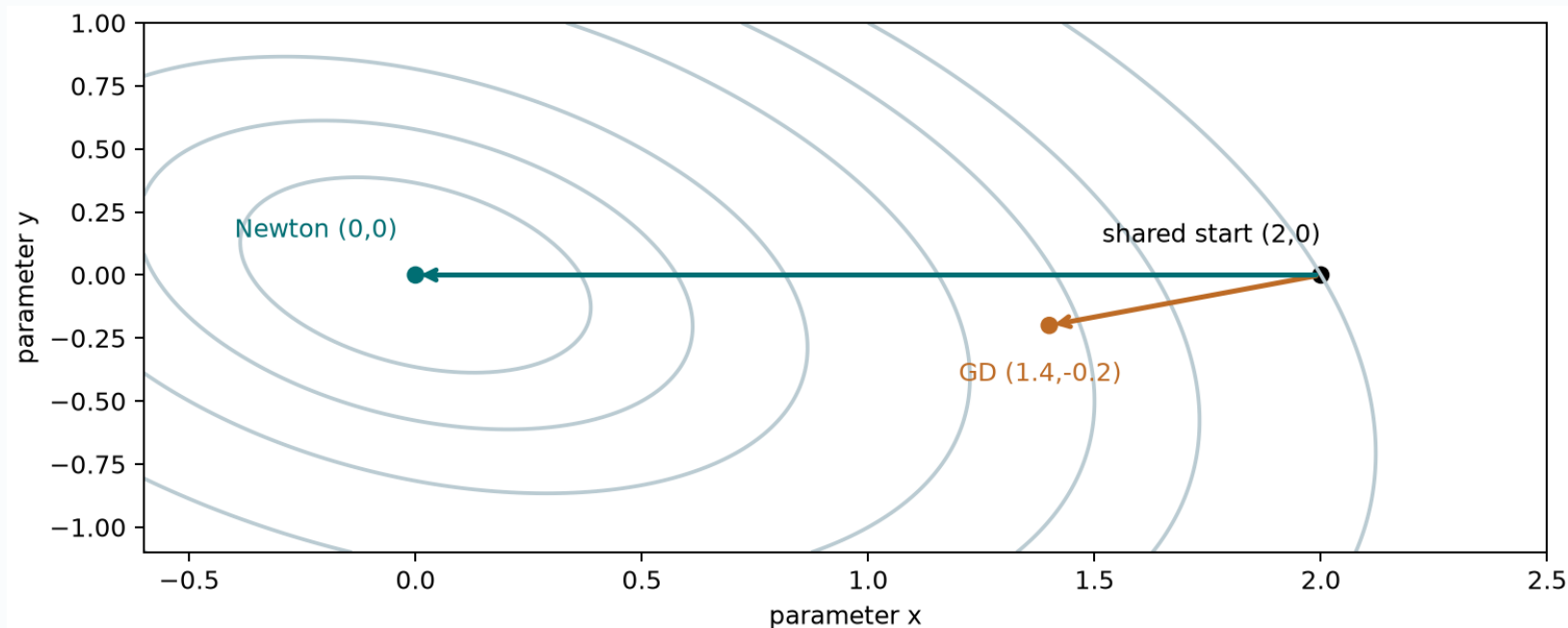
method	starting loss	new point	new loss
GD rate0.1	6	(1.4, -0.2)	2.72
full Newton	6	(0, 0)	0

Both lower this quadratic. Their directions differ because Newton accounts for the mixed curvature rather than multiplying every gradient coordinate by the same scalar.

WORKED EXAMPLE

Two alternative arrows from one starting point

Contours connect points with equal loss.



The arrows share the original point (2,0). The gradient-descent point is not an intermediate stage of the Newton calculation.

WORKED EXAMPLE

Why the exact quadratic finishes in one full step

The local quadratic model is the entire function in this special case.

For an exact positive-definite quadratic and an exact solve, a full Newton step reaches the unique minimum from any starting point. General neural-network losses do not have this guarantee.

$$\eta_N = 0.5 : \quad w_{new} = (2, 0) - 0.5(2, 0) = (1, 0)$$

$$E(1, 0) = 1.5(1)^2 = 1.5$$

A damped Newton step keeps the learning rate below one here. It moves only partway along the correction, so the one-step result no longer applies.

The Hessian can also be expensive, singular or indefinite. These issues are separate from the arithmetic of a successful small quadratic example.

WORKED EXAMPLE

An invertible Hessian can lead to a saddle

Newton seeks a stationary point; that point need not be a minimum.

$$S(x, y) = \frac{x^2 - y^2}{2}, \quad w_0 = (0, 1), \quad g_0 = (0, -1)$$

H is diagonal (1,-1), which is invertible but indefinite. Solve $Hs=g$: $sx=0$ and $-sy=-1$, so $s=(0,1)$.

$$w_{new} = (0, 1) - (0, 1) = (0, 0)$$

$$S_{old} = -0.5, \quad S_{new} = 0$$

The loss increases and the origin is a saddle. Reaching the exact stationary point in one step is not the same as reaching a minimum.

WORKED EXAMPLE

A singular Hessian makes the raw formula undefined

Even a true minimum may have zero curvature at that point.

$$T(w) = w^4, \quad g(w) = 4w^3, \quad H(w) = 12w^2$$

$$w = 0 : \quad g = 0, \quad H = 0, \quad g/H = 0/0 \text{ undefined}$$

The origin is a minimum because w -to-the-fourth is nonnegative. But the raw Newton division cannot be evaluated there. Do not call $0/0$ a zero correction.

A practical method needs an appropriate stopping test or a modified step when the curvature system cannot be solved.

INDEPENDENT PRACTICE

Your turn: a fresh mixed quadratic

Reset to the stated starting point for each alternative method.

$$Q(x, y) = 2x^2 + xy + 2y^2, \quad w_0 = (1, -1)$$

Calculate the gradient and Hessian. Find the initial loss. Solve $Hs=g$ by elimination and calculate a full Newton step and its loss.

Separately take one gradient-descent step with rate 0.1 from the original start and calculate its loss. Explain correction s versus signed displacement $-s$.

WORKED ANSWER

Practice: ingredients and elimination

Differentiate, substitute the start, then solve the two equations.

$$g = (4x + y, x + 4y) \Rightarrow g_0 = (3, -3)$$

H has rows (4,1) and (1,4), from differentiating those gradient components.

$$Q(1, -1) = 2 - 1 + 2 = 3$$

$$4s_x + s_y = 3, \quad s_x + 4s_y = -3$$

$$s_y = 3 - 4s_x \Rightarrow s_x + 4(3 - 4s_x) = -3$$

$$-15s_x = -15 \Rightarrow s_x = 1, \quad s_y = 3 - 4 = -1$$

WORKED ANSWER

Practice: both updates and both losses

The full Newton correction is $s=(1,-1)$; its displacement is $(-1,1)$.

$$w_N = (1, -1) - (1, -1) = (0, 0), \quad Q_N = 0$$

$$w_G = (1, -1) - 0.1(3, -3) = (0.7, -0.7)$$

$$Q_G = 2(0.7)^2 + (0.7)(-0.7) + 2(-0.7)^2$$
$$= 0.98 - 0.49 + 0.98 = 1.47$$

Common errors: using H inverse entry-by-entry, evaluating H and g at different points, adding s when it was defined by $Hs=g$, dropping the lecture rate without declaring a full step, or assuming invertible means positive definite.